

**Assignment-01**  
**CSE 323**

1. Write down the advantages of compiler over interpreter and also explain about tokens, lexeme and pattern with examples.
2. The compiler operates in various phases; each phase transforms the source program from one representation to another. Every phase takes inputs from its previous stage and feeds its output to the next phase of the compiler. Now make an expression and construct the different compilation phases for that expression and also mention the tasks that are completed by the compiler for each phase.
3. Perform left factoring of the following grammar by showing all the necessary steps -

$\langle \text{sentence} \rangle \rightarrow \langle \text{noun phrase} \rangle \langle \text{verb phrase} \rangle$

$\langle \text{noun phrase} \rangle \rightarrow \langle \text{adjective} \rangle \langle \text{noun phrase} \rangle \mid \langle \text{adjective} \rangle \langle \text{singular noun} \rangle$

$\langle \text{verb phrase} \rangle \rightarrow \langle \text{singular verb} \rangle \langle \text{adverb} \rangle$

4. Write down the major differences between LR and LF.
5. Explain with examples how left recursion and common prefix is a problem in top down parser.
6. Find first and follow function for the following grammar

$S \rightarrow A B C \mid BCf \mid abc$

$A \rightarrow a A \mid \epsilon \mid ASD$

$B \rightarrow b \mid C d \mid SDT$

$C \rightarrow c \mid A e \mid \epsilon$

7. Build an LL(1) parsing table from Q (6) of the modified grammar. Is this an LL(1) grammar? Justify your answer.